```
from google.colab import drive
drive.mount('/content/drive/')
```

```
Mounted at /content/drive/
```

```
!pip install spectral
```

```
Looking in indexes: https://pypi.org/simple, https://us-python.pkg.dev/colab-wheels/public/simple/
Collecting spectral
  Downloading spectral-0.23.1-py3-none-any.whl (212 kB)
     ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 212.9/212.9 KB 13.1 MB/s eta 0:00:00
Requirement already satisfied: numpy in /usr/local/lib/python3.9/dist-packages (from spectral) (1.22.4)
Installing collected packages: spectral
Successfully installed spectral-0.23.1
```

```
%tensorflow_version 2.x
import tensorflow as tf
import random
from tensorflow.keras import Sequential,layers
from tensorflow.keras.models import Model
from tensorflow.keras.optimizers import Adam
from tensorflow.keras.callbacks import ModelCheckpoint
import statistics
from sklearn.decomposition import PCA
from sklearn.model_selection import train_test_split
from sklearn.metrics import confusion_matrix, accuracy_score, classification_report, cohen_kappa_score
import tensorflow_probability as tfp
from operator import truediv
from tensorflow.compat.v1.distributions import Bernoulli
from plotly.offline import init_notebook_mode

import numpy as np
import matplotlib.pyplot as plt
import scipy.io as sio
import os
import spectral
import tensorflow as tf
from tensorflow.python.keras import backend as K

init_notebook_mode(connected=True)
%matplotlib inline
```

```
Colab only includes TensorFlow 2.x; %tensorflow_version has no effect.
```

```
# Global variables
im_width, im_height, im_depth, im_channel = 11,11,30, 1
mc_loss_weight = 5
```

Model Construction

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Data Loading

[ ] ↳ *5 cells hidden*

Prepare Training and validation Dataset

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Training

[ ] ↳ *4 cells hidden*

Tuning_5

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Testing

```
def test_episode(test_patches_class,test_class_labels,test_C,test_K,i,f) :
  selected_classes = test_class_labels[i:f]   # [1, 2, 3, 4, 5, 6, 7, 8]
  support_labels = list(selected_classes)
  query_labels = []
  support_patches = []
  query_patches = []
  for x in selected_classes :
    y = test_class_labels.index(x)
    support_imgs = test_patches_class[y][:test_K,:,:,:,:]
    query_imgs = test_patches_class[y][test_K:,:,:,:,:]
    support_patches.extend(support_imgs)
    query_patches.extend(query_imgs)
    for i in range(query_imgs.shape[0]) :
      query_labels.append(x)
  temp1 = list(zip(query_patches, query_labels))
  random.shuffle(temp1)
  query_patches, query_labels = zip(*temp1)
  x = len(query_labels)
  query_patches = tf.convert_to_tensor(np.reshape(np.asarray(query_patches),(x,im_height,im_width,im_depth,1)),dtype=tf.float32)
  support_patches = tf.convert_to_tensor(np.reshape(np.asarray(support_patches),(test_C*test_K,im_height,im_width,im_depth,1)),dtype
  return query_patches, support_patches, query_labels, support_labels,x
```

```
query_patches, support_patches, query_labels, support_labels,x = test_episode(test_patches_class,test_class_labels,3,5,0,3)
```

```
# Metrics to gather
test_loss = tf.metrics.Mean(name='test_loss')
test_acc = tf.metrics.Mean(name='test_accuracy')
def test_step(support, query, support_labels, query_labels, K, C, y):
    loss, mc_predictions, mean_accuracy, classwise_mean_acc, y = ProtoModel(support, query, support_labels, query_labels, K, C, y,n_
    return loss, mc_predictions, mean_accuracy, classwise_mean_acc, y

```

```
for epoch in range(1): #1000
    test_loss.reset_states()
    test_acc.reset_states()
    tquery_patches1, tsupport_patches1, query_labels1, support_labels1, x1 = test_episode(test_patches_class,test_class_labels,3,5,0
    loss1, mc_predictions1, mean_accuracy1, classwise_mean_acc1, y1 = test_step(tsupport_patches1, tquery_patches1,support_labels1,
    print('OA1',mean_accuracy1)
# Class-wise Accuracy
    for i in range(tC) :
      print('class',i+1,classwise_mean_acc1[i])
    print('loss',loss1)
    tquery_patches2, tsupport_patches2, query_labels2, support_labels2, x2 = test_episode(test_patches_class,test_class_labels,3,5,3
    loss2, mc_predictions2, mean_accuracy2, classwise_mean_acc2, y2 = test_step(tsupport_patches2, tquery_patches2,support_labels2,
    print('OA2',mean_accuracy2)
# Class-wise Accuracy
    for i in range(tC) :
      print('class',i+4,classwise_mean_acc2[i])
    print('loss',loss2)
```

```
Temperature Scaling Predictions :  tf.Tensor([0. 0. 1.], shape=(3,), dtype=float32)

Loop Number :  8
AAAAAAAAAAAAAAAAAAAA
Dists :  tf.Tensor([474.41028   507.4191      2.8871663], shape=(3,), dtype=float32)
BBBBBBBBBBBBBBBBBBBB
New Distance (T applied):  tf.Tensor([431.28207   461.29007     2.6246965], shape=(3,), dtype=float32)
CCCCCCCCCCCCCCCCCCCC
Softmax Predictions :  tf.Tensor([0. 0. 1.], shape=(3,), dtype=float32)
DDDDDDDDDDDDDDDDDDDD
Temperature Scaling Predictions :  tf.Tensor([0. 0. 1.], shape=(3,), dtype=float32)

Loop Number :  9
AAAAAAAAAAAAAAAAAAAA
Dists :  tf.Tensor([372.3883   409.9993    13.791505], shape=(3,), dtype=float32)
BBBBBBBBBBBBBBBBBBBB
New Distance (T applied):  tf.Tensor([338.53482  372.72662   12.537731], shape=(3,), dtype=float32)
CCCCCCCCCCCCCCCCCCCC
Softmax Predictions :  tf.Tensor([0. 0. 1.], shape=(3,), dtype=float32)
DDDDDDDDDDDDDDDDDDDD
Temperature Scaling Predictions :  tf.Tensor([0. 0. 1.], shape=(3,), dtype=float32)

OA2 tf.Tensor(0.95049506, shape=(), dtype=float32)
class 4 [1.0]
class 5 [0.925]
class 6 [1.0]
loss tf.Tensor(113.16363, shape=(), dtype=float32)
```

Overall Accuracy

```
mean_predictions1 =  tf.reduce_mean(mc_predictions1,axis=0)
mean_predictions2 =  tf.reduce_mean(mc_predictions2,axis=0)
overall_predictions = tf.concat([mean_predictions1,mean_predictions2],axis=0)
overall_true_labels = tf.concat([y1,y2],axis=0)
correct_pred = tf.cast(tf.equal(                                        # accuracy for the current pass
            tf.cast(tf.argmax(overall_predictions, axis=-1), tf.int32),
            tf.cast(tf.argmax(overall_true_labels,axis=-1), tf.int32)), tf.float32)
o_acc = tf.reduce_mean(correct_pred)
print("Overall accuracy:",o_acc)
```

```
Overall accuracy: tf.Tensor(0.88814694, shape=(), dtype=float32)
```

Confusion Matrix

```
from sklearn.metrics import confusion_matrix
from sklearn.metrics import accuracy_score
from sklearn.metrics import classification_report
mean_predictions1 =  tf.reduce_mean(mc_predictions1,axis=0)
cm_pred1 = tf.argmax(mean_predictions1, axis=-1)
mean_predictions2 =  tf.reduce_mean(mc_predictions2,axis=0)
cm_pred2 = tf.argmax(mean_predictions2, axis=-1) + 3
overall_predictions = tf.concat([cm_pred1,cm_pred2],axis=0)
cm_true1 = tf.argmax(y1,axis=-1)
cm_true2 = tf.argmax(y2,axis=-1) + 3
overall_true_labels = tf.concat([cm_true1,cm_true2],axis=0)
results = confusion_matrix(overall_true_labels,overall_predictions)
print ('Confusion Matrix :')
print(results)
print ('Report : ')
print (classification_report(overall_true_labels, overall_predictions))
```

```
Confusion Matrix :
[[ 41   0   0   0   0   0]
 [ 52 180   0   0   0   0]
 [  0   0  23   0   0   0]
 [  0   0   0  15   0   0]
 [  0   0   0  11 185   4]
 [  0   0   0   0   0  88]]
Report :
              precision    recall  f1-score   support

           0       0.44      1.00      0.61        41
           1       1.00      0.78      0.87       232
           2       1.00      1.00      1.00        23
           3       0.58      1.00      0.73        15
           4       1.00      0.93      0.96       200
           5       0.96      1.00      0.98        88

    accuracy                           0.89       599
   macro avg       0.83      0.95      0.86       599
weighted avg       0.94      0.89      0.90       599
```

Kappa Accuracy

```
import sklearn
sklearn.metrics.cohen_kappa_score(overall_true_labels, overall_predictions)
```

```
0.8498089539056858
```

✓ 0s completed at 12:49 PM